

Report

Project 6

By

Vimlesh Sonawane

Introduction to Analytics

ALY6000

Professor Wada Roy



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1. Introduction

The purpose of this project was to explore and apply fundamental statistical concepts using R programming. The focus was on probability distributions such as binomial, Poisson, and normal. In addition, this assignment required the use of simulations, data exploration, visual representation, and analysis of real-world datasets, including the palmerpenguins dataset. This process enhanced my understanding of both theoretical principles and practical applications of statistics. It also helped me to work effectively with R functions used for density, probability, quantiles, and random number generation.

Some of the most valuable insights from the project include:

- The binomial distribution clearly illustrated how the probability of winning a certain number of games is shaped by underlying probabilities.
- Poisson distribution modeling effectively represented the frequency of calls in a typical call center, helping identify how staffing affects the likelihood of meeting service quotas.
- Analysis of the normal distribution provided valuable context for understanding product life cycles, especially in identifying defective light bulbs based on their lifespan.
- Visualization techniques, including histograms and density plots, confirmed theoretical principles such as the Central Limit Theorem by displaying how sample means cluster around population averages.
- Exploring the palmerpenguins dataset revealed interesting biological patterns. In particular, the data showed a positive relationship between flipper length and beak depth in Gentoo penguins and a symmetric distribution of flipper length in Adelie penguins.

Visuals and tables were especially helpful in supporting these findings:

- Histograms confirmed expected distributions.
- Scatter plots and density curves demonstrated data patterns and variability.
- Tables provided a clear summary of statistics across different species and genders.

2. Results and Analysis

The simulation of the Red Sox's performance over a seven-game series showed that the expected number of wins is approximately 4.55, while the chance of exactly five wins is around 0.31. Simulated results aligned well with theoretical expectations.

The Poisson distribution showed that an employee typically handles about seven calls per hour. The probability of a five-person team reaching 275 calls in a day significantly dropped if even one employee was unavailable. This demonstrated how the model can be used to manage operations.

Using the normal distribution, I found that about ninety-five percent of light bulbs are expected to last between 1800 and 2200 hours. Additionally, light bulbs falling below the lower ten percent threshold (approximately 1872 hours) were classified as defective.

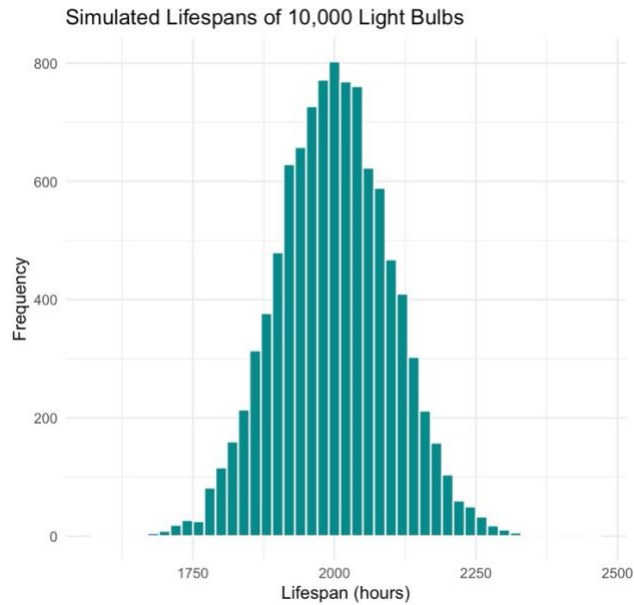
Simulating one thousand sample means from a normally distributed population confirmed that they centered closely around the true mean of 2000 hours. Both histogram and density plots supported the Central Limit Theorem, showing a tight and normal spread of values.

When analyzing penguin data, I observed that the flipper length in Adelie penguins follows a near-normal distribution, while in Gentoo penguins, flipper length and beak depth showed a strong positive correlation, suggesting a biological link between these features.

3. Visual Summary

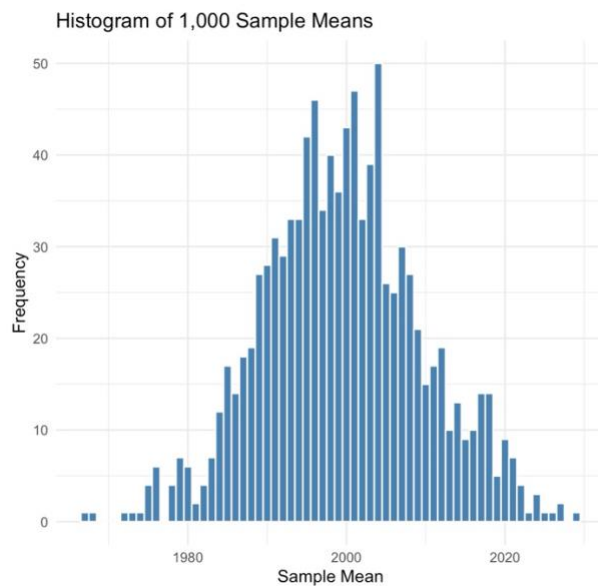
3.1. Visualizations and their insights:

- **Histogram of Light Bulb Lifespan:**



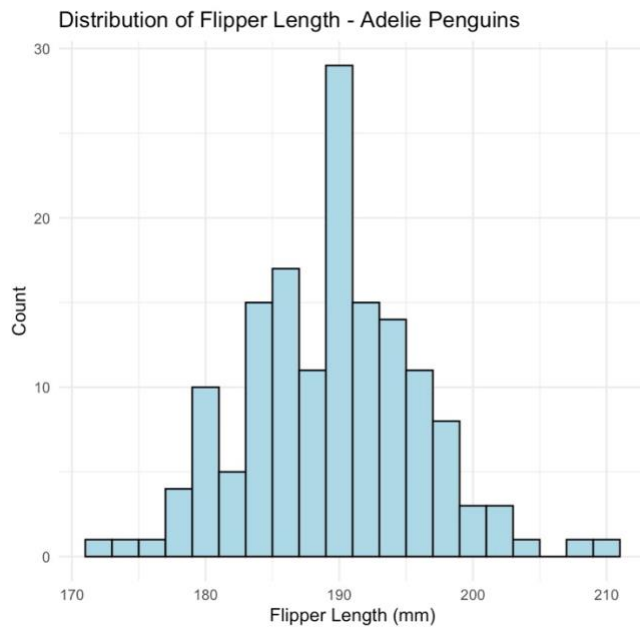
Displayed a symmetrical bell curve centered at 2000 hours, supporting the assumption of normality. This visualization helped define the range in which most bulbs are expected to function and determine the threshold for defects.

- **Histogram of Sample Means:**



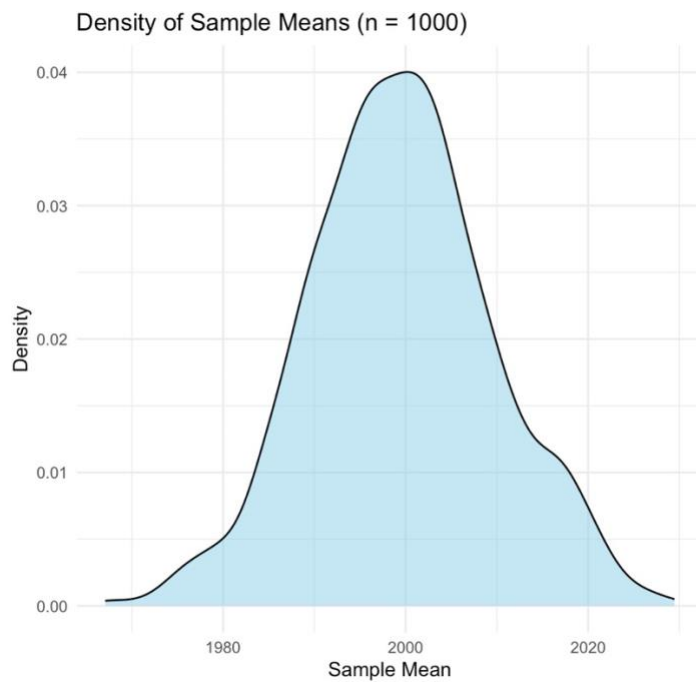
Illustrated that most means gathered from repeated samples were closely centered around the population mean. This reinforced the idea that sample averages are reliable predictors of population behavior.

- **Density Plot of Sample Means:**



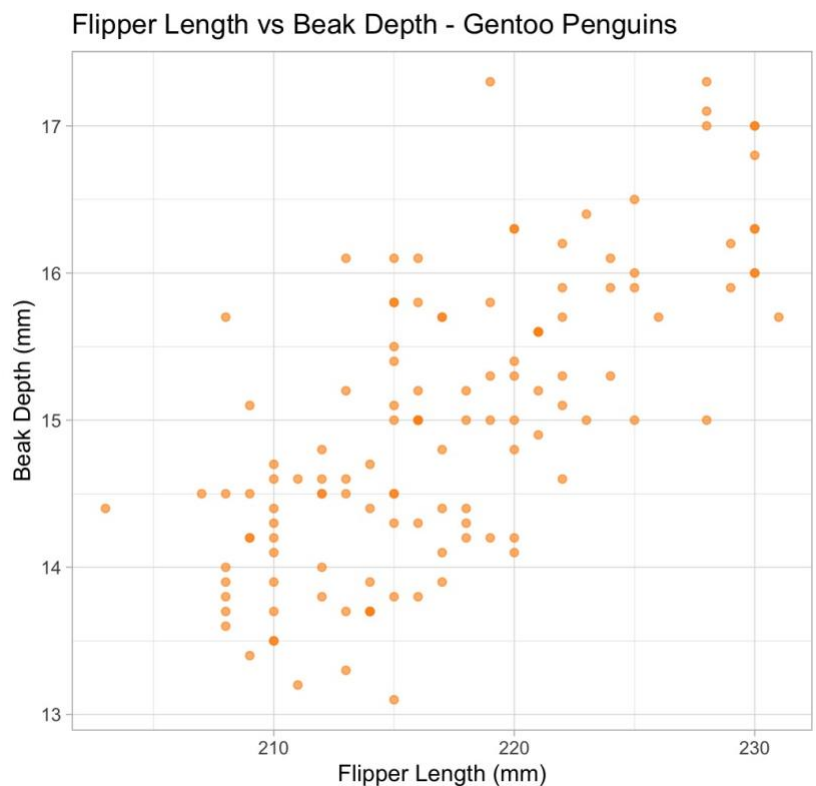
Offered a smoother and continuous look at the sample mean distribution. It further confirmed that the means consistently cluster around the expected value.

- **Histogram of Adelie Penguin Flipper Length:**



Revealed a balanced and unimodal shape. It visually supported the assumption that the flipper lengths in this group follow a normal distribution.

- Scatter Plot of Gentoo Penguins:



Showed a clear and consistent upward trend, confirming that longer flippers in Gentoo penguins are often paired with deeper beaks. This suggested a meaningful correlation between the two measurements.

3.2. Tables and their insights:

- Table 1 (Summary for Gentoo Penguins):

Table: Summary Statistics – Gentoo Penguins

mean_flipper	mean_beak_depth	mean_body_mass
217.187	14.98211	5076.016

Displayed the average measurements for flipper length, beak depth, and body mass. These metrics provided context for analyzing traits and comparing them to other species.

- **Table 2 (Species Counts):**

Table: Penguin Counts by Species

```
|species | count|
|:-----|-----:|
|Adelie  |    152|
|Chinstrap |    68|
|Gentoo  |   124|
```

Presented the total number of observations for each penguin species. It helped evaluate the dataset's completeness and balance.

- **Table 3 (Body Mass by Species and Sex):**

Table: Average Body Mass by Species and Sex

```
|species |sex    | avg_mass|
|:-----|:-----|-----:|
|Adelie  |female | 3368.836|
|Adelie  |male   | 4043.493|
|Adelie  |NA     | 3540.000|
|Chinstrap |female | 3527.206|
|Chinstrap |male   | 3938.971|
|Gentoo   |female | 4679.741|
|Gentoo   |male   | 5484.836|
|Gentoo   |NA     | 4587.500|
```

Gave a breakdown of average body mass based on species and sex. This allowed for the identification of biological trends and potential gender-based differences.

4. Conclusion and Recommendations

This project helped me gain confidence in using R to analyze and visualize data through various probability distributions. It demonstrated how statistical methods apply directly to real-world questions, such as predicting outcomes, planning operations, and evaluating product reliability.

Going forward, I suggest applying these techniques to more real-time datasets, such as live sensor data or streaming sports statistics, to increase the relevance of these tools. Adding bootstrap methods and confidence intervals can further enhance the reliability of conclusions, especially when working with smaller or variable datasets.

References

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Horst, A. M., Hill, A. P., & Gorman, K. B. (2020). *palmerpenguins: Data for Palmer Archipelago (Antarctica) penguins*. R package version 0.1.0. <https://allisonhorst.github.io/palmerpenguins/>